

Exercise 2

1. The characteristic for an n- and p-channel transistor, with the minimum dimensions $W/L=0.8/0.5$, is shown in Figure 1.

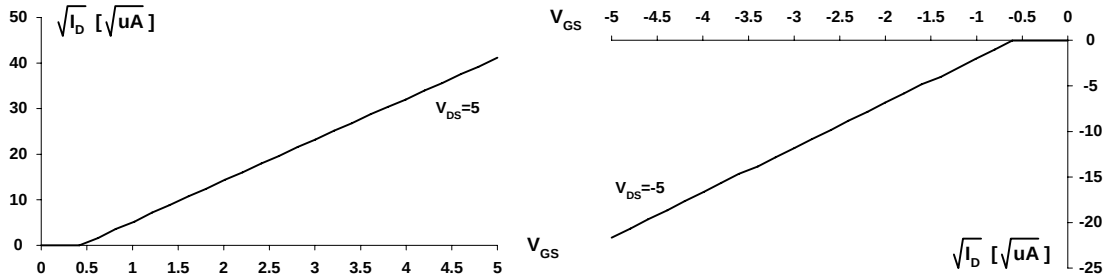


Figure 1: Characteristics of an n- and p-channel transistor.

1a. Calculate the gate threshold voltage (V_M) for an inverter with $W_n/L_n=1.0$ and $W_p/L_p=5.0$. Assume $V_{DD} = 5V$ and disregard from the channel-length modulation.

1b. If the rise and fall times are $t_r=120ps$ and $t_f=100ps$, how much energy (E_{dp}) is dissipated during a switch due to the direct path between ground and V_{DD} ?

1c. How wide must the p-channel transistor be for the gate threshold to be $V_M = 2.5V$? You can assume that $V_{Tn}=|V_{Tp}|$.

2. The characteristic for an n- and p-channel transistor, with the minimum dimensions $W/L=0.5/0.35$, is shown in Figure 2.

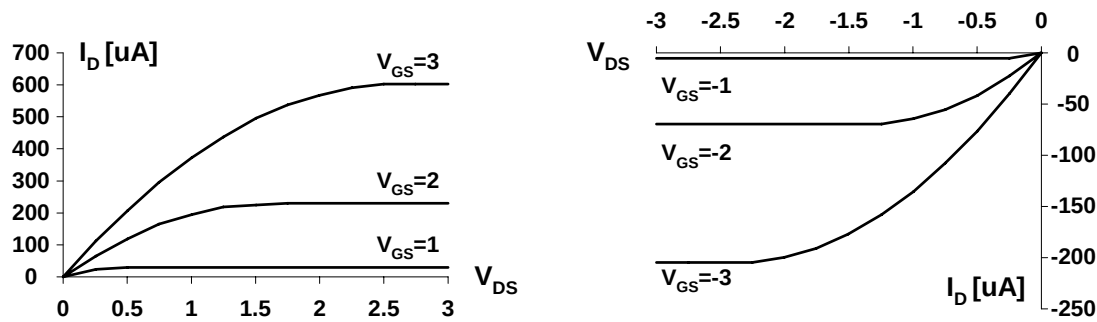


Figure 2: Characteristics of an n- and p-channel transistor.

2a. Design an inverter, i.e. find W_p/L_p if $W_n/L_n=0.5/0.35$, with the gate threshold voltage $V_M = V_{DD}/2 = 1.5V$. Assume $V_{Tn}=|V_{Tp}|=0.5V$ and disregard from the channel-length modulation.

2b. Calculate the gate threshold voltage (V_M), with the dimensions from (a) and threshold voltages (V_{Tn} and V_{Tp}) extracted from Figure 2. Observe that the transconductance (k') calculated in (a) can not be used.

3. Calculate V_{OH} and V_{OL} for the circuit in Figure 3.

$V_{IH} = 3V$, $V_{IL} = 0V$, $V_{DD} = 3V$, $R_{on-n} = 2K$ och $V_{Tn} = 0.5$.

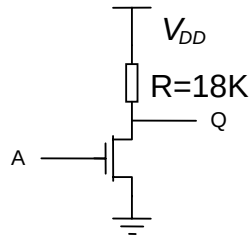


Figure 3: A NMOS-circuit.

4. The characteristic for an n- and p-channel transistor, with the minimum dimensions $W/L=0.5/0.35$ and $V_{tn} = 0.4$ and $|V_{tp}| = 0.6$ is shown in Figure 4.

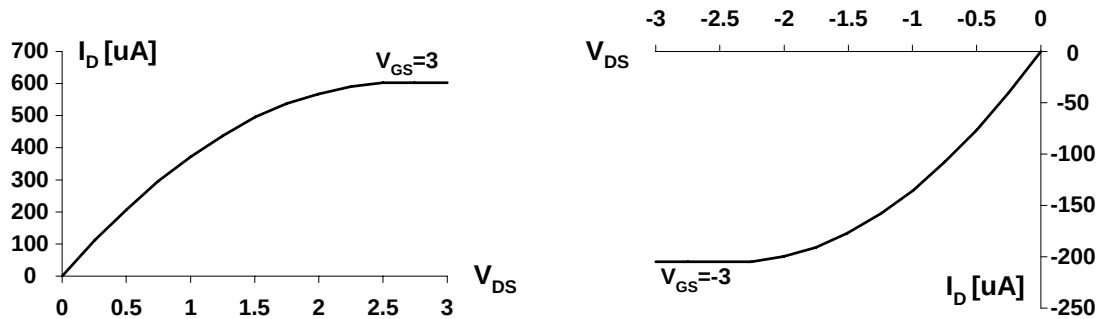


Figure 2: Characteristics of an n- and p-channel transistor.

We want to create a simple RC-model, as shown in Figure 5, of an inverter with $W_n/L_n = 0.5/0.35$ and $W_p/L_p = 1.0/0.35$.

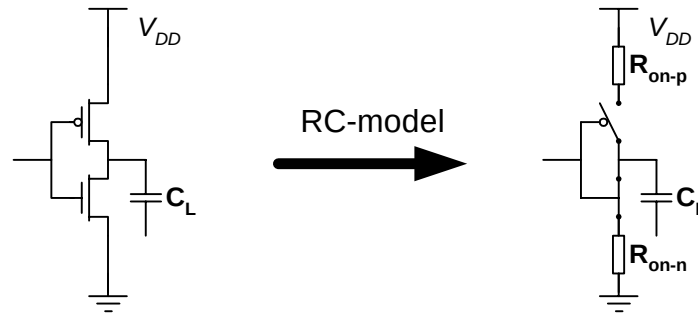


Figure 5: RC-model of a CMOS inverter

A common method to estimate the resistance is to take the mean of the resistance in the saturated region and the linear region according to

$$R_{on-n} = \frac{R_{n(V_{out}=V_{dd})} + R_{n(V_{out}=V_{dd}/2)}}{2}$$

4a. Use the RC-model to calculate the propagation, rise and fall time, assume that $C_L = 100\text{fF}$ and $V_{DD} = 3\text{V}$.

4b. Calculate the propagation time if the physical-model is used.

$$t_p = \frac{CV_{DD}}{k(V_{DD} - V_T)^2}$$

4c. Neglect the threshold voltage. How long is the propagation time?

4d. Short-channel transistors often have linear dependency between the drain and gate voltage. How is the propagation time affected by the linear dependency? Neglect the threshold voltage.

5

5a. Figure 6 shows the voltage transfer function for two different inverters. Why is one of the inverters non-regenerative? Estimate V_{OH} , V_{OL} , V_{IL} and V_{IH} for the regenerative inverter.

5b. Three regenerative inverters from a) are serially connected. What is the output voltage of the third gate assuming 3.25 V input at the 1st gate?

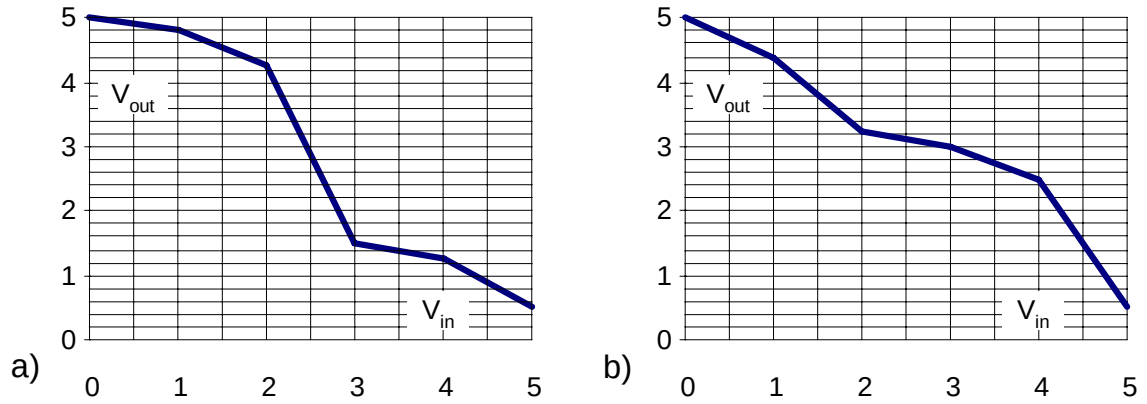


Figure 6: Voltage transfer characteristic (VTC) for two inverters

6. Figure 7 shows an inverter using a pull-up resistor and an NMOS transistor.

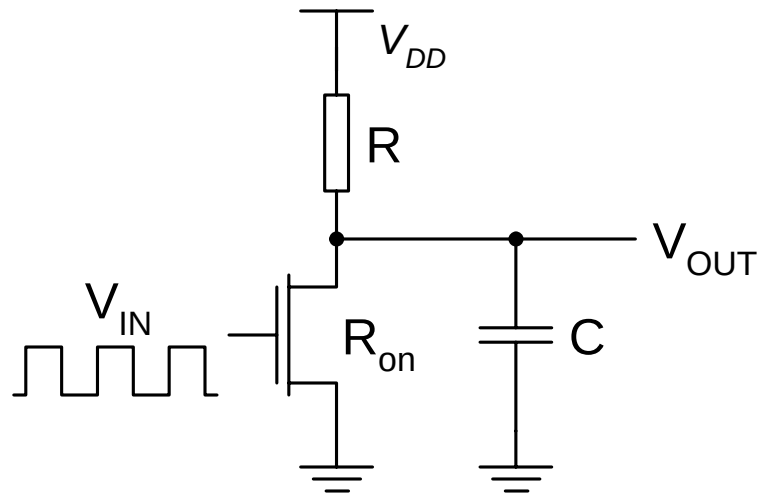


Figure 7: An inverter with resistor pull-up.

6a. What happens with the delays t_{pHL} and t_{pLH} when the resistor R is changed to a resistor $2R$. **Motivate!**

6b. What happens with the output voltages V_{OH} and the inverter threshold V_M when the value of the resistor R is increased. **Motivate!**

6c. Determine the output voltage V_{OL} , the fall time t_f , and the rise time t_r , using the following values $R=35k\Omega$, $R_{on}=5k\Omega$, $V_{DD}=3V$, and $C=45fF$.

6d. What happens with the dynamic power consumption when the resistor R , in Figure 7, is changed to a resistor $2R$. **Motivate!**

6e. Where is the dynamic power dissipated (consumed)?

7. The characteristics for an n- respectively a p-channel transistor is shown in Figure 8. The plots are from a 0.18-micron process and both transistors have a width (W) at 0.6 micron. The supply voltage is 1.8 volt, which is regarded as low. The short-channel effects are thus neglected. The channel length modulation is neglected as well.

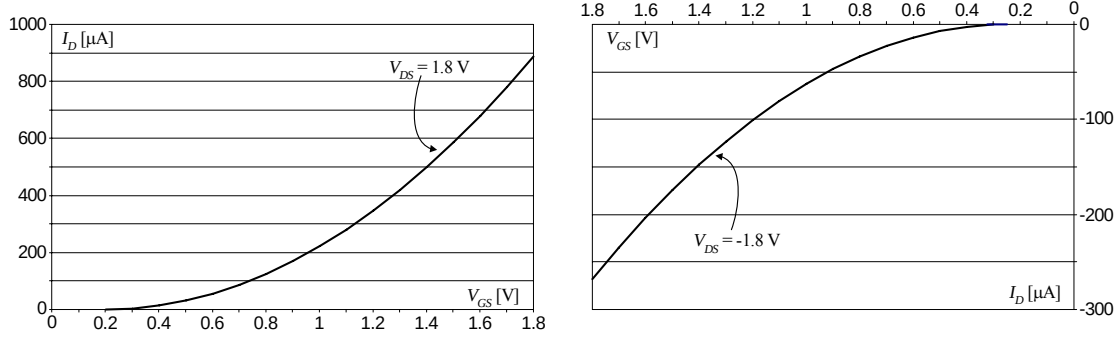


Figure 8: Drain current vs. gate voltage, for a drain-source voltage at 1.8 volt.

7a. Determine the switching threshold voltage (V_M) for a two input NOR-gate with the transistor widths $W_n=1.2$ respectively $W_p=2.4$ microns.

7b. Calculate the maximum direct-path current in the NOR-gate.

7c. Determine the width of the p-channel transistors if the switching threshold voltage (V_M) is at half the supply voltage, i.e. when $V_M = 0.9V$ (assume that $V_{Tn}=|V_{Tp}|$).

8.

Figure 9a shows a network consisting of 7 metal wires, where each wire has the length $L = 20$ mm. Figure 9b introduces a simple model for each wire. The metal has the sheet resistance $R_{\square}=0.07 \Omega$, the bottom plate capacitance $C_b=0.04 \text{ fF}/\mu\text{m}^2$ and the fringe capacitance $C_f=0.06 \text{ fF}/\mu\text{m}$ towards the substrate.

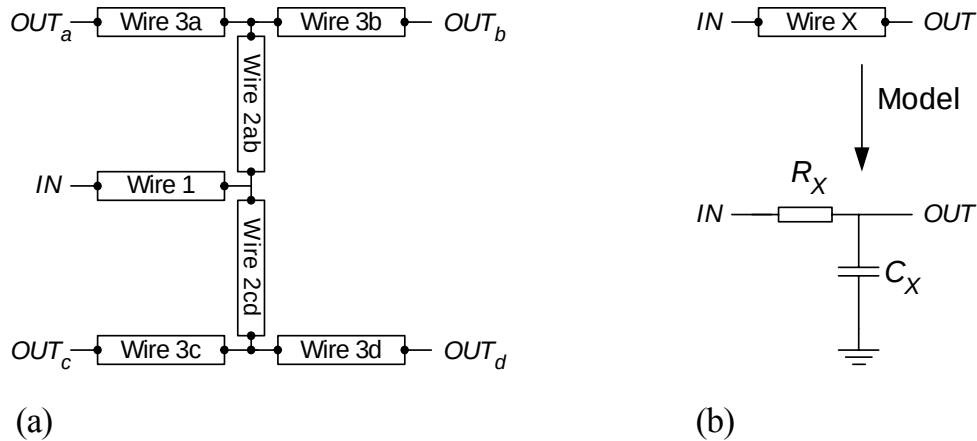


Figure 9: A distribution network and a wire model for each wire

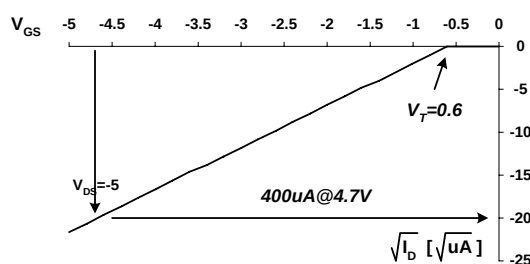
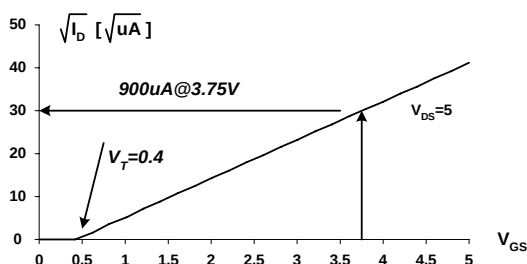
8a. Assume that each wire has a width of $W = 1.0 \mu\text{m}$. Use the lumped RC model and determine the propagation delay time at the outputs OUT_i . Assume an ideal step at the input IN .

8b. Use the Elmore delay equation to determine the propagation delay time for the whole network at the outputs OUT_i .

8c. The first wire feeds all other wires. The current is thus higher in the beginning wires compared to the wires at the end. A practiced method is to use wider wires in the beginning to reduce the current density. Therefore, it would be interesting to see how that affects the propagation delay. Assume that Wire 1 has the width $W_1 = 4.0 \mu\text{m}$, Wire 2_i has the width $W_2 = 2.0 \mu\text{m}$ and Wire 3_i has the width $W_3 = 1.0 \mu\text{m}$. Use the Elmore delay equation to determine the propagation delay time for the whole network at the outputs OUT_i .

Solution 2

1.



1a

From the Figure:

$W_n=0.8$, $W_p=0.8$, $L_n=0.5$, $L_p=0.5$, $V_{DD}=5$

$V_{Tn}=0.4$, $V_{Tp}=0.6$

Extract two points from the Figure to use in equation 3.29 page 93. For example:

n-transistor: $V_{GS}=3.75V$, $I_D=900e-6$ A

p-transistor: $V_{GS}=4.7V$, $I_D=400e-6$ A

$$I_{Dn}=kn'/2 \cdot W_n/L_n \cdot (3.75 - V_{Tn})^2 = 900e-6 \Rightarrow kn' = 100e-6$$

$$I_{Dp}=kp'/2 \cdot W_p/L_p \cdot (4.7 - V_{Tp})^2 = 400e-6 \Rightarrow kp' = 30e-6$$

For the inverter

$W_n/L_n=1$ and $W_p/L_p=5$

$kp=kn' \cdot 5.0 = 150e-6$

$kn=kn' \cdot 1.0 = 100e-6$

$$r = \sqrt{kp/kn} \Rightarrow r = 1.22 \quad (\text{Page 186})$$

$$V_M = (r \cdot (V_{DD} - V_{Tp}) + V_{Tn}) / (1 + r) \Rightarrow V_M = 2.60 \text{ V} \quad (\text{Eq. 5.6, page 186})$$

1b

$$I_{DnM} = kn/2 \cdot (V_M - V_{Tn})^2 \Rightarrow I_{DnM} = 242e-6 \text{ A}$$

$$I_{DpM} = kp/2 \cdot (V_M - V_{DD} + V_{Tp})^2 \Rightarrow I_{DpM} = 242e-6 \text{ A}$$

$$E_{dp} = V_{DD} \cdot I_{DnM} \cdot (t_r + t_f) / 2 = 5 \cdot 242e-6 \cdot 110e-12 = 121e-15 \text{ J}$$

1c

$$V_M = \frac{V_{DD}}{2}, \quad V_{Tn} = |V_{Tp}|, \quad \text{long channel} \Rightarrow r = 1$$

$$r = 1 = \sqrt{k_n / |k_p|} \Rightarrow \frac{W_p}{L} |k_p'| = \frac{W_n}{L} k_n' \Rightarrow W_p = \frac{W_n}{L} \frac{k_n'}{|k_p'|} L = 1 \times \frac{100}{30} \times 0.5 = 1.67$$

2a.

$$V_M = (V_{Tn} + r(V_{DD} + V_{Tp})) / (1+r) = V_{DD}/2 \text{ with } r = \sqrt{-k_p/k_n} \text{ and } V_{Tn} = |V_{Tp}| \Rightarrow$$

$$r = 1 \text{ and } -k_p = k_n$$

$$k_p = W_p/L_p * k_p'$$

$$k_n = W_n/L_n * k_n'$$

Extract two points from Figure 2 to use in equation 3.29 page 93. For example:

$$\text{n-transistor: } V_{GS}=3V, I_D=600e-6 A$$

$$\text{p-transistor: } V_{GS}=3V, I_D=200e-6 A$$

$$I_{Dn} = k_n' / 2 * W_n / L_n * (3 - V_{Tn})^2 = 600e-6 \Rightarrow k_n' = 134e-6$$

$$I_{Dp} = k_p' / 2 * W_p / L_p * (3 - V_{Tp})^2 = 200e-6 \Rightarrow k_p' = 45e-6$$

Observe that $W_p/L_p = W_n/L_n = 0.5/0.35$ in Figure 2.

$$W_p/L_p = W_n/L_n * k_n' / -k_p' = 0.5/0.35 * 2.98 = 1.49/0.35$$

2b.

In Figure 2:

$$W_n=0.5, W_p=0.5, L_n=0.35, L_p=0.35$$

Extract two points in each plot in the saturated area with different V_{GS} .

$$I_1 = k_n' / 2 * W_n / L_n * (3 - V_{Tn})^2 = 600$$

$$I_2 = k_n' / 2 * W_n / L_n * (2 - V_{Tn})^2 = 230$$

Gives two solutions

$$V_{Tn} = 0.374 \text{ and } k_n' = 121.85 \text{ or}$$

$$V_{Tn} = 2.382 \text{ and } k_n' = 2202,$$

were the first one is correct.

$$I_3 = k_p' / 2 * W_p / L_p * (3 - V_{Tp})^2 = 205$$

$$I_4 = k_p' / 2 * W_p / L_p * (2 - V_{Tp})^2 = 70$$

Gives two solutions

$$V_{Tp} = 0.594 \text{ and } k_p' = 49.58 \text{ or}$$

$$V_{Tp} = 2.369 \text{ and } k_p' = 720,$$

were the first one is correct.

Alternative: Extract V_{Tp} and V_{Tn} graphical by plotting V_{GS} against $\sqrt{I_D}$ were the values are taken from the saturated region of Figure 2. With V_{Tp} and V_{Tn} , k' can be found the same way as in 2a.

For the inverter:

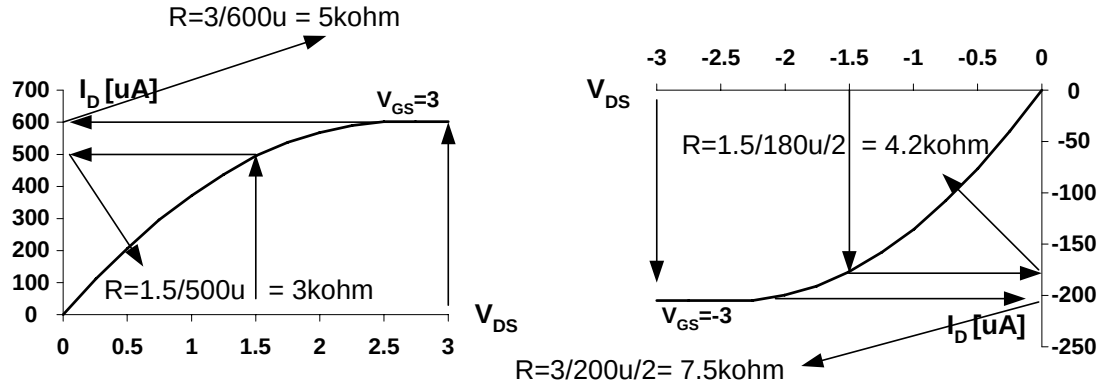
$$W_p=1.49, W_n=0.5$$

$$r = \sqrt{W_p * k_p' / W_n / k_n'} = 1.06$$

$$V_M := (r * (3 - 0.59) + 0.37) / (1 + r) = 1.42$$

3. $V_{OL}=0.3V$ (voltage division between R and R_{on-n}) and $V_{OH} = 3V$.

4a.



$R_{VDDn} = 3/600e-6 = 5k\Omega$ and $R_{VDD/2n} = 1.5/500e-6 = 3k\Omega$
 $R_{VDDp} = 3/(2*200e-6) = 7.5k\Omega$ and $R_{VDD/2p} = 1.5/(2*180e-6) = 4.2k\Omega$
 The factor 2 used to calculate R_{VDDp} and $R_{VDD/2p}$ is due to $Wp=1.0$
 $R_{on-n} = 4.0k\Omega$ and $R_{on-p} = 5.8k\Omega$

$t_r = 2.2*5.8k*100f = 1.28ns$. (Top of page 30 and Figure 1-19 page 27)
 $t_f = 2.2*4k*100f = 0.88ns$.

$t_{plh} = 0.69*5.8k*100f = 0.40ns$. (Eq. 5.19, 5.18, 5.20, page 200)
 $t_{phl} = 0.69*4.0k*100f = 0.28ns$.
 $t_p = (0.4+0.28)/2 = 0.34ns$

4b.

$Wn/Ln = Wp/Ln = 0.5/0.35$, $V_{DD} = 3$, $V_{Tn} = 0.4$, $V_{Tp} = 0.6$, $V_{GS} = 3$
 $I_{Dn} = kn'/2 * Wn/Ln * (V_{GS} - V_{Tn})^2 = 600 \Rightarrow Kn' = 124.3e-6$
 $I_{Dp} = kp'/2 * Wp/Lp * (V_{GS} - V_{Tp})^2 = 200 \Rightarrow Kp' = 48.6e-6$

$Kn = 0.5/0.35 * kn' = 177.5e-6$
 $Kp = 1/0.35 * kp' = 138.9e-6$

$t_{plh} = 100e-15 * 3 / (139e-6 * (V_{DD} - V_{Tp})^2) = 0.375ns$
 $t_{phl} = 100e-15 * 3 / (178e-6 * (V_{DD} - V_{Tn})^2) = 0.249ns$
 $t_p = (t_{plh} + t_{phl}) / 2 = 0.312ns$

4c.

$t_{plh} = 100e-15 / (139e-6 * V_{DD}) = 0.240ns$
 $t_{phl} = 100e-15 / (178e-6 * V_{DD}) = 0.187ns$
 $t_p = (t_{plh} + t_{phl}) / 2 = 0.214ns$

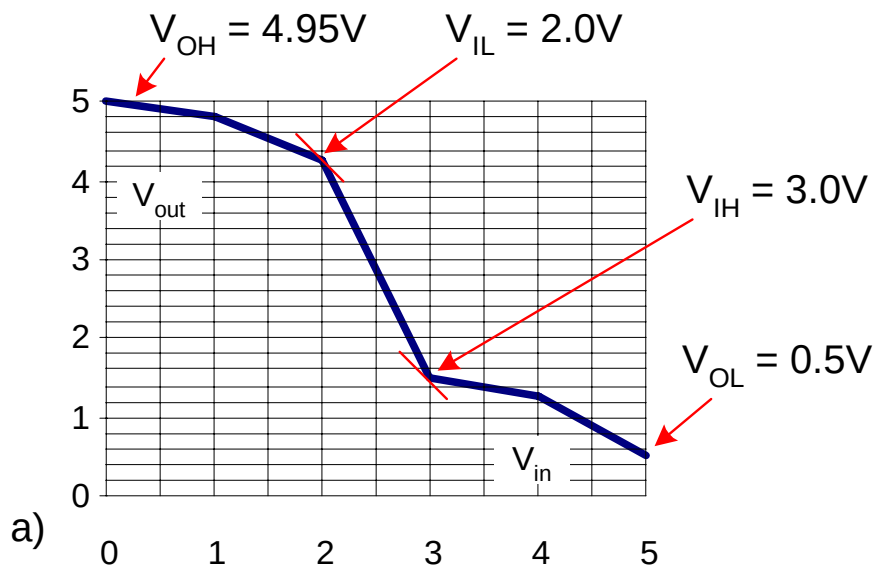
4d.

$$t_{plh} = 100e-15 / 139e-6 = 0.719ns$$

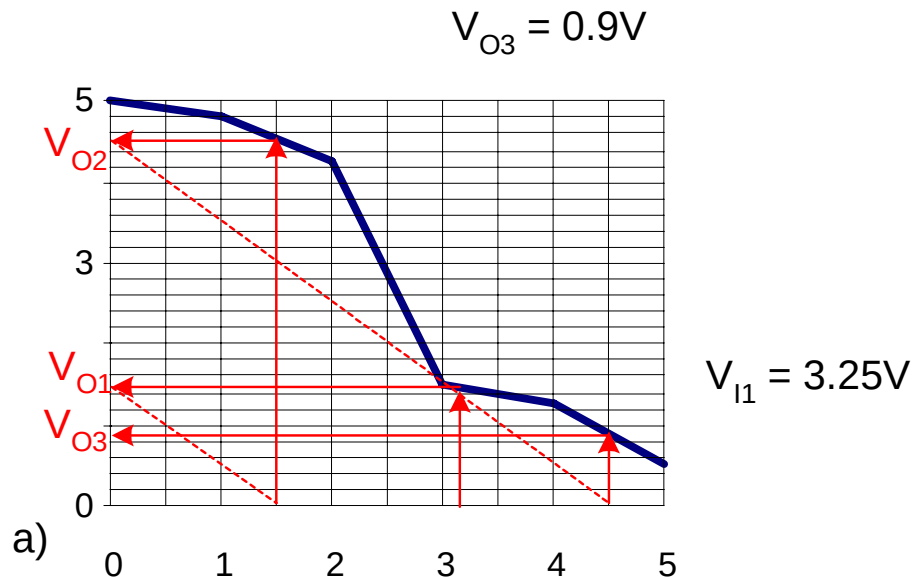
$$t_{phl} = 100e-15 / 178e-6 = 0.561ns$$

$$t_p = (t_{plh} + t_{phl}) / 2 = 0.641ns$$

5a. Figure 8b is non-generative since there is no gain in the circuit (Figure 1-15, page 23). V_{OH} is the output value when V_{OL} is applied to the input and vice versa (page 20). To determine the values, V_{OH} and V_{OL} have to be iterated a few times in the graph below (However, 5 V respectively 0.5 V is good enough in this case). V_{IH} and V_{IL} are defined as the input voltages when the VTC slope is -1 (page 21).



5b. The voltages can be be iterated as in Figure 1-15, page 23 in the text book or as in the Figure below. The output after three inverters is thus 0.9 volt.



6a. The pull-down delay, determined by $t_{pHL} = C / (k_n \times V_{DD})$, is not affected by a change of R . The pull-up delay is determined by $t_{pLH} = 0.69RC$. If R is doubled, the delay will also be doubled.

6b. When the transistor is turned off, V_{OH} will reach V_{DD} since the only low impedance path at the output is through the resistor R . The inverter threshold is determined by setting the current through the resistor R equal to the drain current:

$$I_D = (V_{DD} - V_M) / R = k_n (V_M - V_T)^2 / 2$$

Solving V_M , We get:

$$V_M = \sqrt{r(r - 2(V_{DD} + V_T))} - r + V_T \quad \text{where} \quad r = \frac{1}{k_n R}$$

V_M will thus decrease when the resistance increases.

6c. The voltage V_{OL} , is determined by voltage division between the two resistors:

$$V_{OL} = \frac{R_{on}}{R_{on} + R} V_{DD} = \frac{5}{5 + 35} 3 = 0.375V$$

The fall time t_f and the rise time t_r , is determined by (page 29-30):

$$t_f = 2.2 R_{on} C = 2.2 \times 5000 \times 45 \times 10^{-15} = 0.50ns$$

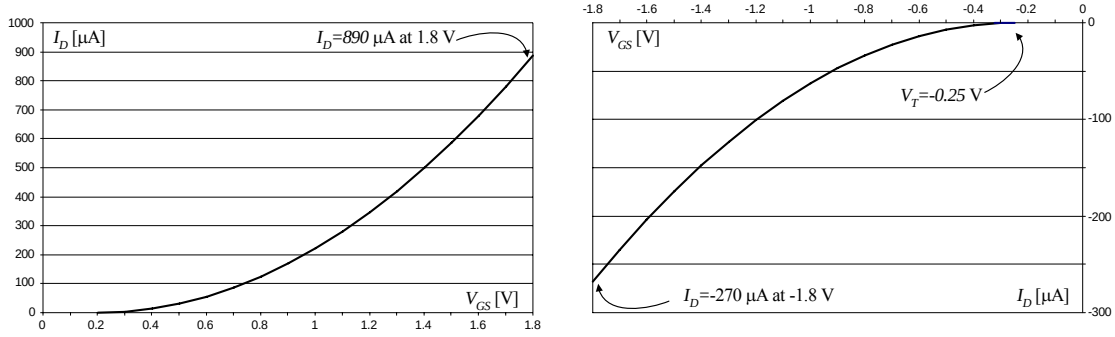
$$t_r = 2.2 R C = 2.2 \times 35000 \times 45 \times 10^{-15} = 3.50ns$$

6d. The dynamic power consumption is determined by $P = f C V_{DD}^2$. There is no dependence of the resistance the power is thus not affected by a change of R .

6e. Half the dynamic power will be lost in R during the charging of C and the second half of the dynamic power will be lost in the transistor (R_{on}) during the pull-down phase. The static power when the transistor is open as well as the short circuit power should also be added. The power is thus lost as heat in the resistances in the components.

7.

a - alternative 1 From the Figure we get the threshold voltages and drain currents.



The following is given for the plots

$$W_n = 0.6 \mu m; \quad W_p = 0.6 \mu m; \quad L_n = 0.18 \mu m; \quad L_p = 0.18 \mu m; \quad V_{DD} = 1.8 V$$

The threshold voltages are given in the plots

$$V_{Tn} = 0.2 V; \quad V_{Tp} = 0.25 V$$

The transistor is in the saturation region. The transconductance are given by solving the drain current equations

$$I_{Dn} = \frac{k'_n}{2} \frac{W_n}{L_n} (V_{DD} - V_{Tn})^2 = 890 \mu A$$

$$k'_n = 209 \mu A/V^2$$

$$I_{Dp} = \frac{k'_p}{2} \frac{W_p}{L_p} (V_{DD} - V_{Tp})^2 = 270 \mu A$$

$$k'_p = 67 \mu A/V^2$$

The NOR-gate use two times wider transistors in the PDN and four times wider transistors in the PUN, which gives the gain factors

$$k_p = k'_p \times 2.4 = 251 \mu A/V^2$$

$$k_n = k'_n \times 1.2 = 161 \mu A/V^2$$

The switching threshold is thus given by (p185-186)

$$r = \sqrt{\frac{\frac{1}{2}k_p}{k_n}} = 0.5662 \quad \left(\frac{1}{2}k_p \text{ since 2 stacked transistors}\right)$$

$$V_M = \frac{r \times (V_{DD} - V_{Tp}) + V_{Tn}}{1 + r} = 0.6880V$$

If both the n-transistors are open (not requested)

$$r = \sqrt{\frac{\frac{1}{2}k_p}{2k_n}} = 0.4004 \quad (2k_n \text{ since 2 parallel transistors})$$

$$V_M = \frac{r \times (V_{DD} - V_{Tp}) + V_{Tn}}{1 + r} = 0.5860V$$

Answer: 0.69 V (0.59 if both n-channel transistors are open)

a - alternative 2 The problem with the alternative 1 is that it is difficult to get a good estimate of the threshold voltage (V_T) from the diagram. An alternative is to plot the square root of the drain current. The threshold voltage is then found when the curve is crossing the x-axis.

a - alternative 3 The switching threshold voltage can also be found by solving the two equations:

$$\begin{cases} I_{Dn1} = \frac{k'_n}{2} \times \frac{W_n}{L_n} \times (1.8 - V_{Tn})^2 = 890 \\ I_{Dn2} = \frac{k'_n}{2} \times \frac{W_n}{L_n} \times (1.4 - V_{Tn})^2 = 500 \end{cases}$$

$$k'_n = 209, V_{Tn} = 0.20$$

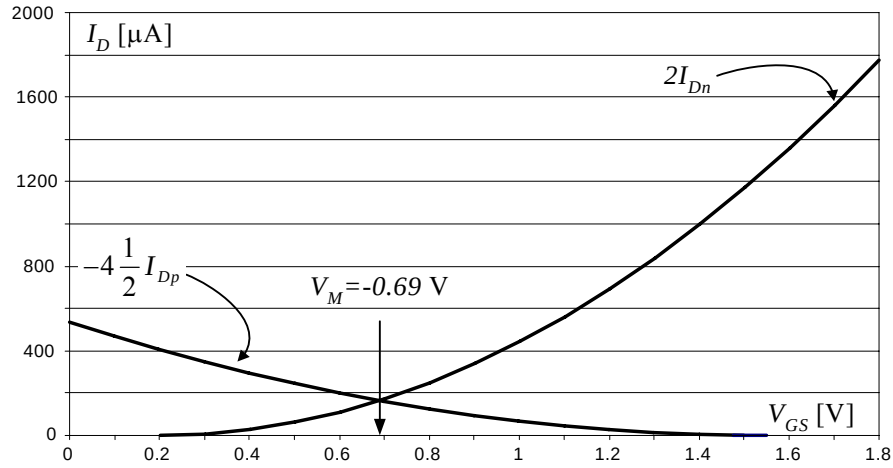
$$k'_n = 10215, V_{Tn} = 1.57 \text{ (not realistic)}$$

$$\begin{cases} I_{Dp1} = \frac{k'_p}{2} \times \frac{W_p}{L_p} \times (1.8 - V_{Tp})^2 = 270 \\ I_{Dp2} = \frac{k'_p}{2} \times \frac{W_p}{L_p} \times (1.4 - V_{Tp})^2 = 150 \end{cases}$$

$$k'_p = 66, V_{Tp} = 0.23$$

$$k'_p = 3084, V_{Tp} = 1.57 \text{ (not realistic)}$$

a - alternative 4 The Switching threshold voltage can be found graphically by plotting both currents in the same diagram. The n-channel current should be double since the transistor width has double size. The p-channel current should be multiplied by four for the same reason and divided by two since there are two stacked transistors.



b The maximum direct path current is given by the switching Threshold voltage. Both transistors are in the saturation region at that point. Both the PDN and the PUN can be thus be used.

$$I_{dp-n} = \frac{k_n}{2} (V_M - V_{Tn})^2 = 29.87 \mu A$$

$$I_{dp-p} = \frac{k_p}{2 \times 2} (V_M - V_{DD} + V_{Tp})^2 = 29.87 \mu A$$

Answer: 30 μA

c Since the threshold voltages are equal, the width of the p-channel transistors can be calculated by

$$W_{p-balance} = \frac{W_n \times k'_n \times 2}{k'_p} = 3.743 \mu m$$

Answer: 3.7 μm

8.

Each wire has the length $L = 20$ mm. The metal has the sheet resistance $R_{\square} = 0.07 \Omega$, the bottom plate capacitance $C_b = 0.04$ fF/ μm^2 and the fringe capacitance $C_f = 0.06$ fF/ μm towards the substrate. The plate capacitor for each wire is

$$C_{xplate} = 20 \times 10^{-3} \times 1 \times 10^{-6} \times 0.04 \times 10^{-15} / 10^{-12} = 0.8 \text{ pF} .$$

The fringing capacitance should be counted twice since there are two fringes. The fringing capacitance is thus

$$C_{xfringe} = 2 \times 20 \times 10^{-3} \times 0.06 \times 10^{-15} / 10^{-6} = 2.4 \text{ pF}$$

The total capacitance will be

$$C_x = 3.2 \text{ pF}$$

A wire which is 2 cm long and 1 μm wide, have

$$20 \times 10^{-3} / 1 \times 10^{-6} = 20000 \text{ } 1 \times 1 \mu\text{m}^2 \text{ squares}$$

Each square has the resistance 0.07 Ω . The total resistance will thus be

$$R_x = 20000 \times 0.07 = 1.4 \text{ k}\Omega$$

8a. In the lumped model we have 3 wires from the input to all outputs. The resistance is thus

$$R = 3 \times R_x = 4.2 \text{ k}\Omega$$

The total capacitance has 7 wires which yields

$$C = 7 \times C_x = 22.4 \text{ pF}$$

The propagation delay is

$$t_p = 0.69RC = 0.69 \times 4.2 \times 10^3 \times 22.4 \times 10^{-12} = 65 \text{ ns}$$

8b. Symmetry gives that we only have to determine the delay to one output e.g. from IN to OUT_a. Equation 4.13, page 154, gives:

$$\begin{aligned} t_p &= 0.69(R_1C_1 + R_1C_{2cd} + R_1C_{3c} + R_1C_{3d} + (R_1 + R_{2ab})C_{2ab} + \\ &+ (R_1 + R_{2ab})C_{3b} + (R_1 + R_{2ab} + R_{3a})C_{2ab}) = \\ &= 0.69 \times 11 \times R_x C_x = 0.69 \times 11 \times 1.4 \times 10^3 \times 3.2 \times 10^{-12} = 34 \text{ ns} \end{aligned}$$

8c. The first wire is 4 times wider and the two second wires are 2 times wider than the four ending wires. The resistances are thus

$$R_1 = \frac{R_x}{4}; \quad R_2 = \frac{R_x}{2}; \quad R_3 = R_x$$

The plate capacitors are determined in a similar way

$$C_{1plate} = 4 \times 0.8 \text{ pF} = 3.2 \text{ pF}; \quad C_{2plate} = 2 \times 0.8 \text{ pF} = 1.6 \text{ pF}; \quad C_{3plate} = 0.8 \text{ pF}$$

The fringing capacitance is however not affected by the widening of the wires. The total capacitances will thus be

$$C_1 = 2.4 + 3.2 = 5.6 \text{ pF}; \quad C_2 = 2.4 + 1.6 \text{ pF} = 4.0 \text{ pF}; \quad C_3 = 2.4 + 0.8 = 3.2 \text{ pF}$$

This yields the propagation delay

$$\begin{aligned} t_p &= 0.69(R_1C_1 + R_1C_2 + R_1C_3 + R_1C_3 + (R_1 + R_2)C_2 + \\ &+ (R_1 + R_2)C_3 + (R_1 + R_2 + R_3)C_2) = \\ &= 0.69\left(\frac{R_x}{4}(C_1 + C_2 + 2C_3) + \left(\frac{R_x}{4} + \frac{R_x}{2}\right)(2C_2 + C_3) + R_xC_2\right) = \\ &= 0.69R_x\left(\frac{1}{4}C_1 + \frac{11}{4}C_2 + \frac{5}{4}C_3\right) = \\ &0.69 \times \frac{1.4 \times 10^3}{4} (5.6 + 11 \times 4.0 + 5 \times 3.2) \times 10^{-12} = \\ &0.69 \times \frac{1.4 \times 10^3}{4} 65.6 \times 10^{-12} = 16 \text{ ns} \end{aligned}$$

The widening of the wires will thus reduce the delay!